Solutions to Problem Set 3

Solution 1. Let d be the greatest common divisor of the numbers $2^{2n+1}-1$ and 2^m+1. Then d is an odd number and

$$2^{2n+1}-1=ad, \qquad 2^m+1=bd,$$

where a and b are positive integers.

So,

$$2^{2n+1}=ad+1, \qquad 2^m=bd-1,$$

and

$$2^{(2n+1)m}=(ad+1)^m=td+1, \qquad 2^{(2n+1)m}=(bd-1)^{2n+1}=ud-1,$$

where t and u are positive integers.

Consequently, since $td+1=ud-1$ we have $d(u-t)=2$. That is, $d \mid 2$. But d is an odd number, therefore $d=1$ and we are done.

Solution 2. Since $\mathcal{A}=pr$, (p represents the semi-perimeter of $\triangle ABC$), then $\mathcal{A}-ra=rp-ra=r(p-a)$ (cyclic) and the given expression can be written as

$$\begin{aligned}
&\frac{\cos A-\cos B}{r(p-c)}+\frac{\cos B-\cos C}{r(p-a)}+\frac{\cos C-\cos A}{r(p-b)}\\
&=\frac{1}{r}\left(\frac{\cos A-\cos B}{p-c}+\frac{\cos B-\cos C}{p-a}+\frac{\cos C-\cos A}{p-b}\right)\\
&=\frac{2}{r}\left(\frac{\cos A-\cos B}{a+b-c}+\frac{\cos B-\cos C}{b+c-a}+\frac{\cos C-\cos A}{a-b+c}\right)
\end{aligned}$$

Applying the Law of Cosine, yields

$$\begin{aligned}
\frac{\cos A-\cos B}{a+b-c}&=\frac{1}{a+b-c}\left(\frac{c^2+b^2-a^2}{2bc}-\frac{c^2+a^2-b^2}{2ac}\right)\\
&=\frac{b(c^2+b^2-a^2)-a(c^2+a^2-b^2)}{2abc(a+b-c)}\\
&=\frac{(a-b)[c^2-(a+b)^2]}{2abc(a+b-c)}=\frac{p(b-a)}{abc}
\end{aligned}$$

Likewise, $\frac{\cos B-\cos C}{b+c-a}=\frac{p(c-b)}{abc}$ and $\frac{\cos C-\cos A}{a-b+c}=\frac{p(a-c)}{abc}$. Hence,

$$\begin{aligned}
&\frac{\cos A-\cos B}{\mathcal{A}-rc}+\frac{\cos B-\cos C}{\mathcal{A}-ra}+\frac{\cos C-\cos A}{\mathcal{A}-rb}\\
&=\frac{2}{r}\left(\frac{\cos A-\cos B}{a+b-c}+\frac{\cos B-\cos C}{b+c-a}+\frac{\cos C-\cos A}{a-b+c}\right)\\
&=\frac{2}{r}\left(\frac{p(b-a)}{abc}+\frac{p(c-b)}{abc}+\frac{p(a-c)}{abc}\right)=0
\end{aligned}$$

and we are done.

Solution 3. Number of integer solutions of $x+y+z=31$ with $x\in\{0,\ldots,7\}$, $y\in\{0,2,4,6,8,10\}$, $z\in\{0,5,10,15,20,25,30\}$ equals the coefficient of x^{31} in $(1+x+\cdots+x^7)(1+x^2+\cdots+x^{10})(1+x^5+\cdots+x^{30})$, which is 10.

Solution 4. Since $ad=bc$ can be thought of as $\frac{a}{b}=\frac{c}{d}=r$, write $a=rb$, $c=rd$. Then

$$\frac{\sqrt{a}-\sqrt{b}}{\sqrt{c}-\sqrt{d}}=\frac{\sqrt{r}-1}{\sqrt{r}-1}\sqrt{\frac{b}{d}}=\sqrt{\frac{b}{d}}=\frac{\sqrt{b}\sqrt{r-1}}{\sqrt{d}\sqrt{r-1}}=\sqrt{\frac{a-b}{c-d}}.$$

Solution 5. Since $6x-y^2-7$, $6y-z^2-7$ and $6z-x^2-7$ are indices of roots, then

$$\begin{aligned}6x-y^2-7&\geq 2,\\6y-z^2-7&\geq 2,\\6z-x^2-7&\geq 2.\end{aligned}$$

Adding the preceding inequalities, yields

$$x^2+y^2+z^2-6(x+y+z)+27\leq 0$$

or

$$(x-3)^2+(y-3)^2+(z-3)^2\leq 0$$

which only solution is $x=y=z=3$. Substituting these values in the statement, we get

$$\sqrt[6x-y^2-7]{\sqrt[6y-z^2-7]{\sqrt[6z-x^2-7]{\sqrt{728x+729y+730z}}}}=\sqrt{\sqrt{\sqrt{6561}}}=3$$

and we are done.

Remark. Considering the counting number 1 we have the system

$$\begin{aligned}6x-y^2-7&=1,\\6y-z^2-7&=1,\\6z-x^2-7&=1,\end{aligned}$$

which solutions are $(2,2,2)$ and $(4,4,4)$. In both cases the values of the expression $728x+729y+730z$ are 4374 and 8748 respectively and they are positive integers. But when the index is 1 there is not radicals and surely this case should not be considered.

□

Solution 6. Let $S(n) = \sum_{k=0}^{n} \frac{1}{2^k}\binom{n+k}{n} = \sum_{k=0}^{n} \frac{1}{2^k}\binom{n+k}{k}$. We have, $S(1) = 2$, and

$$\begin{aligned} S(n+1) &= \sum_{k=0}^{n+1} \frac{1}{2^k}\binom{n+1+k}{k} = \sum_{k=0}^{n+1} \frac{1}{2^k}\binom{n+k}{k} + \sum_{k=0}^{n+1} \frac{1}{2^k}\binom{n+k}{k-1} \\ &= S(n) + \frac{1}{2^{n+1}}\binom{2n+1}{n+1} \\ &\quad + \frac{1}{2}\sum_{k=1}^{n+2} \frac{1}{2^{k-1}}\binom{n+1+k-1}{k-1} - \binom{2n+2}{n+1}\frac{1}{2^{n+2}} \\ &= S(n) + \frac{1}{2}S(n+1) \end{aligned}$$

Therefore, $S(n+1) = 2S(n)$ for all $n \geq 1$ and $S(1) = 2$. Then, $S(n) = 2^n$, and

$$\sum_{k=1}^{n} \frac{1}{2^k}\binom{n+k}{n} = 2^n - 1$$

□